

Packet Tracer – ACL Basics and IP Traffic Filtering

Objectives

Part 1: Build a Routed Network Topology

Part 2: Configure IP Addressing

Part 3: Verify Initial Connectivity

Part 4: Configure Standard ACLs

Part 5: Configure Extended ACLs

Part 6: Verify Traffic Filtering

Background / Scenario

In real-world networks, administrators often need to control which devices can communicate with specific networks or services. Access Control Lists (ACLs) provide a method for filtering traffic and enforcing security policies.

In this lab, you will configure both **Standard ACLs** and **Extended ACLs**. You will learn how ACLs can permit or deny traffic based on source IP addresses, destination IP addresses, and protocols.

This lab focuses on:

- Configuring Standard ACLs
- Configuring Extended ACLs
- Applying ACLs to router interfaces
- Verifying ACL functionality
- Understanding ACL placement rules

You are learning one of the most important security features used in Cisco networks.

Instructions

Part 1: Build the Network Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. Add the following devices:

- 1 Router (1941 or 2911)
- 2 Switches (2960)
- 2 PCs

c. Rename the devices:

- Router0 → **R1**
 - Switch0 → **S1**
 - Switch1 → **S2**
 - PC0 → **PC-A**
 - PC1 → **PC-B**
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Step 2: Connect the Devices

Use **Copper Straight-Through** cables.

Device Interface		Device Interface	
PC-A	FastEthernet0	S1	Fa0/1
S1	Gi0/1	R1	Gi0/0
R1	Gi0/1	S2	Gi0/1
S2	Fa0/1	PC-B	FastEthernet0

Wait until all links become green.

Network Addressing Plan

Device Interface IP Address

PC-A	Fa0	192.168.1.10 /24
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Device Interface IP Address

R1	Gi0/0	192.168.1.1 /24
R1	Gi0/1	192.168.2.1 /24
PC-B	Fa0	192.168.2.10 /24

Part 2: Configure IP Addressing

Step 1: Configure PC-A

Navigate to:

Desktop → IP Configuration

Enter:

- IP Address: 192.168.1.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
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Step 2: Configure PC-B

Enter:

- IP Address: 192.168.2.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.2.1
-

Step 3: Configure Router Interfaces

On R1:

enable

configure terminal

interface g0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface g0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

Part 3: Verify Initial Connectivity

Before creating ACLs, verify that communication works normally.

Step 1: Ping from PC-A to PC-B

On PC-A:

ping 192.168.2.10

Expected Result:

Reply from 192.168.2.10

You should receive four successful replies.

Step 2: Ping from PC-B to PC-A

On PC-B:

ping 192.168.1.10

This should also succeed.

Part 4: Configure a Standard ACL

What is a Standard ACL?

A Standard ACL filters traffic based only on the **source IP address**.

Standard ACLs use numbers:

1 - 99

1300 - 1999

Best practice:

Place Standard ACLs as close as possible to the destination.

Step 1: Block PC-A from Reaching Network 192.168.2.0

On R1:

```
access-list 10 deny 192.168.1.10
access-list 10 permit any
```

Step 2: Apply the ACL

Apply the ACL outbound toward PC-B.

```
interface g0/1
ip access-group 10 out
exit
```

Step 3: Test Connectivity

From PC-A:

```
ping 192.168.2.10
```

Expected Result:

Request timed out

PC-A should now be blocked.

Step 4: Test Another Host

If another device existed in the 192.168.1.0 network, it would still be allowed because only PC-A was denied.

Part 5: Remove the Standard ACL

Step 1: Remove ACL from Interface

```
interface g0/1
no ip access-group 10 out
exit
```

Step 2: Delete ACL

```
no access-list 10
```

Connectivity should now be restored.

Part 6: Configure an Extended ACL

What is an Extended ACL?

Extended ACLs can filter using:

- Source IP
- Destination IP
- Protocol
- Port Numbers

Extended ACLs use numbers:

100 - 199

2000 - 2699

Best practice:

Place Extended ACLs as close as possible to the source.

Step 1: Create an Extended ACL

Block ICMP (Ping) traffic from PC-A to PC-B.

```
access-list 100 deny icmp host 192.168.1.10 host 192.168.2.10
access-list 100 permit ip any any
```

Step 2: Apply the ACL

Apply inbound on R1's LAN interface.

```
interface g0/0
ip access-group 100 in
exit
```

Understanding the ACL

```
deny icmp host 192.168.1.10 host 192.168.2.10
```

Meaning:

- Deny Ping traffic
- From PC-A
- To PC-B

```
permit ip any any
```

Allows all remaining traffic.

Part 7: Verify Extended ACL Operation

Step 1: Ping from PC-A

```
ping 192.168.2.10
```

Expected Result:

Request timed out

The ping should fail.

Step 2: Ping Router Interface

```
ping 192.168.1.1
```

This should still succeed because the ACL only blocks traffic destined for PC-B.

Step 3: View ACL Counters

On R1:

show access-lists

Expected output:

Extended IP access list 100

deny icmp host 192.168.1.10 host 192.168.2.10

permit ip any any

Notice the packet counters increase as traffic matches ACL entries.

Part 8: Verify Interface ACLs

Check where ACLs are applied.

show ip interface g0/0

Expected output:

Inbound access list is 100

This confirms the ACL is active.

Understanding ACL Processing

ACLs are processed:

Top → Bottom

The first matching statement wins.

Example:

access-list 100 deny icmp host 192.168.1.10 host 192.168.2.10

access-list 100 permit ip any any

Traffic is evaluated in order.

Implicit Deny

Every ACL automatically ends with:

deny any

even though it is not visible.

Example:

```
access-list 10 permit 192.168.1.0 0.0.0.255
```

Actually behaves like:

```
access-list 10 permit 192.168.1.0 0.0.0.255  
deny any
```

Always include permit statements when needed.

Troubleshooting Tips

If ACLs do not work:

✓ Verify ACL entries:

```
show access-lists
```

✓ Verify ACL placement:

```
show ip interface
```

✓ Confirm ACL direction:

```
in  
out
```

✓ Verify source and destination IP addresses.

✓ Check interface status:

```
show ip interface brief
```

✓ Remember ACLs are processed top-down.

✓ Watch for the implicit deny rule.

What You Learned

- How Standard ACLs filter by source IP address.
- How Extended ACLs filter by protocol and destination.
- How to apply ACLs to interfaces.

- How ACL processing works.
 - The importance of ACL placement.
 - How to verify ACL operation using Cisco show commands.
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Challenge Extensions (Optional)

Try modifying the lab:

1. Block an entire subnet using a Standard ACL.
2. Create an Extended ACL that blocks only HTTP traffic.

`access-list 101 deny tcp any any eq 80`

3. Permit only PC-A while denying all other hosts.
4. Configure named ACLs.

`ip access-list extended BLOCK_PING`

5. Use Simulation Mode to observe ACL filtering.
6. Combine OSPF and ACLs in a larger topology.